Talk + Say

Product Requirements Document · v1.7

March 5, 2026 · Comprehensive spec for independent implementation

This document is fully self-contained. No source files, transcripts, or prior PRD versions are required to implement or evaluate the application. Every screen, field, widget, interaction, data object, and design decision is documented here.

1. Product Overview

Talk + Say is a two-mode bilingual conversation suite delivered as a single HTML file with no back-end dependencies beyond a public PeerJS signalling server and an optional cloud translation API. The app runs entirely in the browser and persists all data in localStorage.

App	Purpose
Talk	Live P2P bilingual chat. Two people on separate devices connect via a shareable link. Every message is displayed side-by-side in both languages in real time.
Say	Phrasebook and catalog manager. Users build, organise, and review a personal library of bilingual phrases. Phrases created in Talk flow into Say automatically or on demand.

1.1 Core Design Principles

- Zero friction to start: share a link, enter a name, begin talking.
- Language first: the app defaults to showing both languages at all times with no user action required.
- Voice is first-class: STT input and TTS playback are available on every text element where they make sense.
- Conversation pace is protected: advanced features (back-translation verify, clarify requests) are accessible but not visually dominant during a live exchange.
- Single-file, no accounts: the entire application ships as one HTML file; users own their data locally.

2. Architecture Overview

2.1 Technology Stack

Layer	Choice
Runtime	Vanilla JS (ES2020) — no framework, no build step
P2P Transport	PeerJS 1.5.4 over WebRTC (CDN: unpkg.com)
Translation (primary)	Chrome Built-in AI window.translation API (on-device, offline capable)
Translation (fallback)	MyMemory public REST API (cloud, internet required)
TTS	Web Speech API SpeechSynthesisUtterance (built into browser)
STT	Web Speech API SpeechRecognition / webkitSpeechRecognition
Storage	localStorage with try/catch guards (ts_ prefix = Talk; say_ prefix = Say)
Fonts	DM Sans (body) + Lora (headings) via Google Fonts CDN

2.2 Screen Architecture

All screens share a single fixed-position #app container (max-width 480px, centred). Screens are absolutely positioned and toggled via opacity + pointer-events using the .active class. showScreen(id) is the sole navigation primitive.

ID	Element / Action	Behaviour & Rules
`SCR-01`	**Onboarding**	3-slide intro shown once on first launch (localStorage key ts_onboarded). Completing it writes the key and navigates to Home.
`SCR-02`	**Home (Talk tab)**	New session card + session list. Tab bar at bottom switches to Say. Active session auto-loads on return.
`SCR-03`	**Join**	Shown when URL contains ?join=sessionId. Partner enters name + language and joins the named session.
`SCR-04`	**Chat**	Full-screen P2P conversation view. No tab bar. Back arrow returns to Home (session stays alive). ··· menu opens Session overlay.

SCR-05	**Settings**	Language prefs, display name, session list management, storage stats. Back button returns to the screen the user came from (chat or home).
SCR-06	**Say Home**	Phrasebook landing: omni-search, language filter, catalog grid. Tab bar at bottom.
SCR-07	**Catalog**	Cards list for a selected catalog or All Phrases. Search bar, + Add button. Back returns to Say Home.

2.3 Overlay System

Overlays are full-screen dimmed sheets that slide up from the bottom. They sit inside #app at z-index 200. Tapping the dimmed backdrop closes the overlay. All overlays use hidden attribute. The following overlays exist:

ID	Element / Action	Behaviour & Rules
OVL-01	**Session menu (···)**	Shows session info: names, languages, session ID, connection status. Buttons: Close · Share · Settings · End.
OVL-02	**End session**	Confirms session end with two options: "Save phrases & end" (bulk save then end) or "End" (discard). Cancel keeps the session open.
OVL-03	**Clarify request**	Shown when user taps "? Clarify" on a partner message. Quoted text + optional note textarea + Send button.
OVL-04	**Clarify answer**	Shown when user taps "! Respond" on own message after partner requested clarification. Quoted text + answer textarea with Markdown hint + Send button.
OVL-05	**Phrase picker**	Slides up from composer. Search + tag chips + language filter toggle. List of phrases with TTS and Use buttons.
OVL-06	**New phrase**	Form: source text, target text, source lang, target lang, auto-translate button, notes, tags, catalog assignment. Save button.
OVL-07	**New catalog**	Form: name, description, emoji icon grid. Create button.
OVL-08	**Tag picker**	Used from both New Phrase and card detail. Shows all tags as chips with multi-select. Search/create new tag. Done button.

OVL-09	**Confirm dialog**	Generic two-button confirm (Cancel + action). Used for deletes and bulk saves.

3. Data Model

3.1 Session Object (localStorage key: ts_sessions)

ID	Element / Action	Behaviour & Rules
sessionId	**String (ID key)**	Derived from both names: normName(myName)+"-"+normName(partnerName). URL-safe, lowercase, max 49 chars.
isInitiator	**Boolean**	true = user who started the session (PeerJS host role). false = user who joined via link.
myName	**String**	Display name of the local user.
partnerName	**String**	Display name of the remote user. Updated from hello handshake.
myLang	**String**	BCP-47 language code for the local user. Set at session creation or on join. Never overridden by partner.
partnerLang	**String**	BCP-47 language code for the partner. Updated from hello handshake.
pin	**String**	4-digit PIN string if PIN lock enabled; empty string otherwise.
createdAt	**Number**	Unix ms timestamp.
lastActivity	**Number**	Unix ms timestamp of last sent or received message.
lastViewedAt	**Number**	Unix ms timestamp of last time user opened this session. Used to compute unread count.

3.2 Message Object (localStorage key: ts_msg_{sessionId})

ID	Element / Action	Behaviour & Rules
id	**String**	uid() — timestamp base36 + random. Primary key.

from	**String**	"me" | "them"
text	**String**	For from=me: the original text typed/spoken. For from=them: the received translation (already in myLang).
orig	**String**	For from=them: the original text in the partner's language as sent by them.
lang	**String**	The language code of the source text (the language it was written in).
translated	**String|null**	For from=me: the translation into partnerLang. Null while translation is in progress.
status	**String**	"sending" | "sent" | "queued" | "received"
timestamp	**Number**	Unix ms.
clarify	**Object**	{status: "none"|"requested"|"answered", text: string}. Tracks clarification state.
attachment	**Object|null**	{type:"image"|"file", name:string, dataUrl:string} or null.
_backTrans	**String|undefined**	Back-translation result (ephemeral — not sent to partner).
_verdict	**String|undefined**	"good" | "flag" — user's assessment of back-translation quality.

3.3 Card Object (localStorage key: say_cards)

ID	Element / Action	Behaviour & Rules
id	**String**	uid(). Primary key.
catalogIds	**Array<String>**	IDs of catalogs this card belongs to. Empty array = "All Phrases" only.
sourceLang	**String**	BCP-47 code of the source text language.
targetLang	**String**	BCP-47 code of the translation language.
source	**String**	The phrase in the source language.
target	**String**	The translation.

backTranslation	**String**	Back-translation of target back into sourceLang. Empty until Verify is run.
backApproved	**Boolean**	true if user marked back-translation as "Sounds right".
backFlagged	**Boolean**	true if user marked back-translation as "Incorrect". Also adds "incorrect" tag.
notes	**String**	Free-form notes. Supports Markdown: **bold** *italic* ~~strike~~ [link](url).
tags	**Array<String>**	Tag slugs (lowercase, hyphenated). Drawn from global tag registry.
usage	**Number**	Count of times card was used via "Use" button to insert into composer.
createdAt	**Number**	Unix ms.
updatedAt	**Number**	Unix ms.
sourceSessionId	**String|null**	Session ID where phrase was captured, if applicable.
sourcePartnerName	**String|null**	Partner name from originating session, if applicable.

3.4 Catalog Object (localStorage key: say_catalogs)

ID	Element / Action	Behaviour & Rules
id	**String**	"cat-" + uid().
name	**String**	Display name. Max 40 chars.
emoji	**String**	Single emoji character for visual identification.
desc	**String**	Optional description. Max 100 chars.
createdAt	**Number**	Unix ms.

3.5 Preferences Object (localStorage key: ts_prefs)

ID	Element / Action	Behaviour & Rules

`userName`	**String**	Saved display name. Pre-filled into new session and join screens.
`myLang`	**String**	Default source language for new sessions. Default: "en".
`partnerLang`	**String**	Default partner language for new sessions. Default: "th".

4. Flow 1: Live Conversation (Talk)

4.1 Session Setup — Initiator

ID	Element / Action	Behaviour & Rules
`F1.1.1`	**My Name field**	Text input. oninput triggers updateSessionId() and saveMyName(). Pre-filled from prefs.userName. Max 30 chars.
`F1.1.2`	**Partner Name field**	Text input. oninput triggers updateSessionId(). Max 30 chars.
`F1.1.3`	**I Speak selector**	<select> populated with all 21 supported languages. Defaults to prefs.myLang. onchange saves to prefs.
`F1.1.4`	**Partner Speaks selector**	<select> populated with all 21 supported languages. Defaults to prefs.partnerLang. onchange saves to prefs.
`F1.1.5`	**Session ID pill**	Read-only display. Shows normName(myName)+"-"+normName(partnerName). Updates live as names are typed.
`F1.1.6`	**Lock button (🔒)**	Toggles PIN protection. When active, generates a random 4-digit PIN, shows the PIN row, and includes the PIN in shared invite text. Toggling off clears the PIN.
`F1.1.7`	**Share button**	Copies invite URL (?join=sessionId) to clipboard. If PIN is set, appends "PIN: XXXX" to the copied text. No automatic share on Start.
`F1.1.8`	**PIN display row**	Hidden by default. Shows when lock is active: PIN value in monospace + Copy button. Dismissed by × button.
`F1.1.9`	**Start Conversation button**	Disabled until both names have valid normalized forms. On tap: creates session object, saves to

		storage, navigates to Chat. Does NOT auto-share the link.
F1.1.10	**Existing session detection**	If the derived sessionId already exists in storage, quickStart() calls loadSession() to reopen the existing session instead of creating a duplicate.

4.2 Session Setup — Partner (Join Flow)

ID	Element / Action	Behaviour & Rules
F1.2.1	**Join screen trigger**	Shown when the URL contains ?join=sessionId. The session ID is pre-filled in the display card and the manual input.
F1.2.2	**Your Name field**	Text input. Pre-filled from prefs.userName. Required for Join button to enable.
F1.2.3	**I Speak selector**	<select> with all 21 languages. Defaults to prefs.myLang. Sets the joining user's myLang. Critical: this is the only place a joining user can set their language before connecting.
F1.2.4	**PIN input**	Hidden by default. Shown if the session was shared with a PIN. Required field when visible.
F1.2.5	**Join button**	Disabled until name + sessionId are present and sessionId contains a hyphen. On tap: creates session object with isInitiator=false, navigates to Chat, begins P2P connection as joiner.
F1.2.6	**Manual session ID input**	Allows partner to type a session ID manually if they didn't arrive via URL. Typing updates the displayed session ID and enables the Join button.

4.3 P2P Connection Protocol

PeerJS peer IDs are derived as: "talksay_" + sessionId + "_h" (host) or random (joiner).

ID	Element / Action	Behaviour & Rules
F1.3.1	**Host peer setup**	Initiator calls initPeerAsHost(). Creates a Peer with ID talksay_{sessionId}_h. Listens for incoming connections. On connection: calls setupConn().

F1.3.2	**Joiner peer setup**	Joiner calls initPeerAsJoiner(). Creates anonymous Peer. On open: connects to talksay_{sessionId}_h. On connection: calls setupConn().
F1.3.3	**hello handshake**	On conn open, both sides send {type:"hello", name, lang}. Receiving hello updates partnerName + partnerLang in session (never touches myLang). Initiator sends one hello total (guarded by _sentHelloBack flag).
F1.3.4	**Heartbeat**	Every 15 seconds the connected side sends {type:"ping"}. Receiver responds with {type:"pong"}. If conn is not open when heartbeat fires, onConnDrop() is called.
F1.3.5	**Reconnection**	onConnDrop() sets connected=false and schedules reconnect with exponential backoff (2s×attempt, max 20s, max 12 attempts). If initiator: retry as host. If joiner: retry as joiner.
F1.3.6	**Offline queue**	Messages sent while disconnected are pushed to msgQueue. On reconnect, flushQueue() sends them in order and updates their status to "sent".
F1.3.7	**Message wire format**	{type:"msg", id, orig, text, lang, timestamp, attachment?}. orig = sender's original text; text = translation; lang = sender's language.
F1.3.8	**Typing indicator**	{type:"typing", name}. Sent at most once per 2 seconds while user is typing. Clears automatically after 3 seconds of no typing events.

4.4 Chat Screen

4.4.1 Header

ID	Element / Action	Behaviour & Rules
F1.4.1	**Back arrow**	Navigates to Home. Session remains alive; reconnect continues in the background.
F1.4.2	**Session title**	Displays "MyName ↔ PartnerName".
F1.4.3	**Language chip**	Shows "MyLanguage → PartnerLanguage". Turns amber with warning if both languages are the same (misconfiguration guard).

F1.4.4	**Connection dot + label**	Green dot = connected. Grey dot = not connected. Label: "Connected" | "Not connected".
F1.4.5	**Reconnect button**	Appears only when disconnected and a session is active. Tapping manually triggers reconnect and hides the button.
F1.4.6	**··· menu button**	Opens the Session overlay (OVL-01).
F1.4.7	**Engine badge**	Shows "On-device translation" (green) or "Cloud translation" (grey) below the header. Reflects which translation path was used for the last message.

4.4.2 Message Bubbles

Each message is a .msg-row element (class "mine" or "theirs") containing a .bubble-card. Own messages align right with teal background; partner messages align left with paper background. Max width 96%.

ID	Element / Action	Behaviour & Rules
F1.4.8	**Bubble header**	Shows: sender name (bold, 12px) · language pair chip (e.g. "English ↔ Thai") · timestamp. Queued messages show a small amber "queued" badge.
F1.4.9	**Bubble body — two columns**	Side-by-side layout. Left column: source language label + original text + TTS button. Right column: target language label + translated text (shimmer loading animation while pending) + TTS button. A 1px divider separates the columns.
F1.4.10	**TTS buttons**	One per language side. Tap to speak; tap again to stop. Playing state: opacity 1 with .playing class. Uses Web Speech API with locale mapping (e.g. th → th-TH).
F1.4.11	**Attachment zone**	Rendered between body and footer if message has an attachment. Images displayed inline (max-width 100%, rounded corners). Other files shown as a pill with filename.
F1.4.12	**Footer bar**	Persistent bar below body containing: Clarify button (partner messages only; or "! Respond" if partner has requested clarification on own message) · + Save button · ▾ expand toggle.
F1.4.13	**▾ Detail pane**	Collapsed by default. Expanded by ▾ button (rotates to ▴ when open). Contains: Back-translation section

		(own messages only) + Clarify section (if clarify active).
F1.4.14	**Back-translation section**	Own messages only. Shows label "Back-translation" + Verify ↺ button inline. Text area shows "Tap Verify ↺ to check" until verified. After verify: shows back-translated text + ✓ Sounds right / ✗ Flag buttons. After verdict: shows ✓ OK or ⚠ Flagged pill in footer bar and Re-verify ↺ button.
F1.4.15	**? Clarify button (theirs)**	Visible on all partner messages. On tap: opens Clarify Request overlay (OVL-03). After request sent: label changes to "? Clarify" with c-active style. After answer received: label changes to "✓ Clarified" and detail pane auto-expands to show the answer.
F1.4.16	**! Respond button (mine)**	Replaces the hidden clarify anchor on own messages when partner has sent a clarify request. Amber color, prominent. On tap: opens Clarify Answer overlay (OVL-04).
F1.4.17	**+ Save button**	Opens New Phrase overlay (OVL-06) pre-filled with source text, target text, and language codes from this message.
F1.4.18	**Verdict pill**	Appears in footer bar after back-translation verdict. Green "✓ OK" or amber "⚠ Flagged". Removed when Re-verify is tapped.

4.4.3 Composer

ID	Element / Action	Behaviour & Rules
F1.4.19	**Voice (STT) button**	Labelled "Voice" with microphone SVG icon. Tapping activates SpeechRecognition in session.myLang locale. Interim results populate the textarea live. Final result replaces interim. Button turns red (.active) while recording. Hidden if browser lacks SpeechRecognition support.
F1.4.20	**Phrasebook button**	Opens Phrase Picker overlay (OVL-05).
F1.4.21	**Attach button**	Opens file picker. Accepts images, video, PDF, doc, txt. Max 4 MB. Images sent inline; other files sent as a file pill.
F1.4.22	**Message textarea**	Placeholder: "Type or speak…". Auto-resizes up to 110px. Enter sends (shift+Enter = newline). oninput:

		resizes + updates send button + sends typing indicator (throttled to 1 per 2s).
F1.4.23	**Send button**	Circular teal button. Disabled when textarea is empty. On tap: creates message object, appends bubble immediately, clears textarea, triggers translation, sends to partner when translation completes.
F1.4.24	**Offline indicator**	Thin amber bar above composer. Visible when disconnected and session is active. Text: "Offline — messages will send when reconnected".
F1.4.25	**Typing indicator**	Thin bar above offline indicator. Shows "{partnerName} is typing…" for 3 seconds after receiving a typing event.

4.5 Session Management

ID	Element / Action	Behaviour & Rules
F1.5.1	**Session list (Home)**	All saved sessions shown as rows. Each row: avatar initial, "MyName ↔ PartnerName", language pair + time-ago. Badges: unread count (teal), pending clarify count (amber). Row actions: Share (↗), Export (↓ JSON), Delete (×). Tapping row re-opens session.
F1.5.2	**End session**	From ··· menu → End. Shows OVL-02. "Save phrases & end": opens bulk save confirm, then ends session after save completes (or immediately if nothing to save). "End": closes connection, clears state, navigates home.
F1.5.3	**Bulk save**	Filters all session messages for those with both source and translated text. Excludes phrases already in the card library (by source text match). Shows confirm dialog with count. On confirm: saves all as cards to "All Phrases".
F1.5.4	**Export session**	Downloads a JSON file containing the session object and all messages. Filename: talksay-{sessionId}-{timestamp}.json.
F1.5.5	**Session persistence**	Active session ID stored in ts_active. On app load, if ts_active exists and the session is found, the app re-opens the chat screen and attempts to reconnect automatically.

5. Flow 2: Phrasebook (Say)

5.1 Say Home Screen

ID	Element / Action	Behaviour & Rules
F2.1.1	**Omni-search field**	Filters across all cards by source text, target text, and tags. Activates at ≥1 character. Results open inline in the Catalog screen with the search pre-filled.
F2.1.2	**+ Card button**	Opens New Phrase overlay (OVL-06) with no pre-fill.
F2.1.3	**Language filter selector**	Filters the catalog grid to show only catalogs that contain cards with the selected source language. Default "All languages".
F2.1.4	**All Phrases card**	Full-width card at top of grid. Shows total card count (filtered by language if active). Shows "⚠ N need review" badge if any cards have the "incorrect" tag.
F2.1.5	**Catalog cards**	2-column grid. Each shows emoji, name, card count, review badge. Delete × button (top-right; 25% opacity, 80% on hover). Tapping opens the catalog.
F2.1.6	**New Catalog card**	Dashed border card. Tapping opens New Catalog overlay (OVL-07).

5.2 Catalog Screen

ID	Element / Action	Behaviour & Rules
F2.2.1	**Back arrow**	Returns to Say Home, re-renders the catalog grid.
F2.2.2	**Title**	Catalog name or "All Phrases".
F2.2.3	**+ Add button**	Opens New Phrase overlay with current catalog pre-selected.
F2.2.4	**Search bar**	Filters cards within this catalog. Filters by source, target, or tag.
F2.2.5	**Card list**	Sorted by updatedAt descending. Each card is a .pb-card element.

5.3 Phrase Card

ID	Element / Action	Behaviour & Rules

F2.3.1	**Card header**	Source text (14px, medium weight), target text (13px, dim). "⚠ needs review" badge if tagged "incorrect". Tapping header row toggles the detail pane.
F2.3.2	**Use button**	Teal chip in header. Tapping inserts the phrase into the active Talk composer (inserting the source text if sourceLang matches myLang, otherwise target text). If no active session, shows toast "Start a conversation in Talk first". Disabled visual state (opacity 0.35) when no session active.
F2.3.3	**Share button**	Copies "source → target (notes)" to clipboard.
F2.3.4	**Delete button**	Opens Confirm overlay. Permanently removes the card.
F2.3.5	**Tag pills row**	Shows up to 3 tags below the header. "+N" overflow indicator if more than 3.
F2.3.6	**Detail pane**	Hidden by default. Opens when header is tapped. Contains 6 layers (see below).

5.4 Card Detail Layers

Each layer has a label row (showing the layer name and actions) and a content area. Layers are bordered and stacked vertically within the expanded card.

ID	Element / Action	Behaviour & Rules
F2.4.1	**Layer 1: Source**	Shows source text + language label. TTS button. Edit button → shows textarea with Save/Cancel. Saving clears backTranslation, backApproved, backFlagged (translation must be re-verified).
F2.4.2	**Layer 2: Translation**	Shows target text + language label. TTS button. Edit button → shows textarea with Save/Cancel/Re-translate buttons. Re-translate calls translation API using the current textarea content as source. Saving also clears back-translation state.
F2.4.3	**Layer 3: Back-translation check**	Shows back-translation result or "Not checked yet". TTS button (visible after verify). Verify ↺ button triggers translate(target, targetLang, sourceLang). After verify: shows ✓ Sounds right / ✗ Incorrect buttons. After verdict: shows ✓ Verified or ⚠ Flagged text. Re-verify resets verdict state.

F2.4.4	**Layer 4: Notes**	Shows notes (Markdown-rendered). Edit button → textarea with Markdown hint "Supports **bold** *italic* [link](url)". Save/Cancel.
F2.4.5	**Layer 5: Tags**	Shows all card tags as removable pills (# prefix). + Add tag opens Tag Picker overlay. Removing a tag saves immediately.
F2.4.6	**Layer 6: Catalog**	<select> showing all catalogs. Current catalog pre-selected. Changing selection saves immediately. Allows moving a card to a different catalog after creation.
F2.4.7	**Card meta row**	Shows: "Used N× ·" (if usage > 0), "Added Nh ago", "· From Talk with PartnerName" (if sourced from a session).

5.5 New Phrase Overlay (OVL-06)

ID	Element / Action	Behaviour & Rules
F2.5.1	**Source phrase field**	Required. Textarea. TTS button. Language selector (<select>). Defaults to session.myLang or prefs.myLang.
F2.5.2	**Translation field**	Optional. Textarea. TTS button. Language selector (<select>). Defaults to session.partnerLang or prefs.partnerLang. Changing language selector clears the field.
F2.5.3	**Auto-translate button**	Disabled until source has text. Calls translate(source, srcLang, tgtLang). Fills translation field. Shows "Translating…" while in progress.
F2.5.4	**Notes field**	Textarea, 2 rows.
F2.5.5	**Tags field**	Chip display + + Add tag button (opens Tag Picker).
F2.5.6	**Catalog selector**	<select> with "All Phrases" + all named catalogs. Pre-selects current catalog if opened from inside a catalog.
F2.5.7	**Save button**	Disabled until source has text. Creates card object, unshifts to cards array, saves to storage. If currently in Catalog screen, re-renders card list.

5.6 Phrase Picker (OVL-05, from Chat Composer)

ID	Element / Action	Behaviour & Rules
F2.6.1	**Search**	Filters by source, target, or tag text.
F2.6.2	**Tag chips**	Horizontal scroll row. Multi-select filters cards to those having all selected tags.
F2.6.3	**Language filter toggle**	Checkbox labelled "All langs". When unchecked (default in an active session): filters to cards whose language pair matches the current session. When checked: shows all cards. If no session active: shows all cards regardless.
F2.6.4	**No-results helper**	When language filter is active and returns no results: shows message with "Show all" inline link that unchecks the filter.
F2.6.5	**TTS button**	Plays source text of each phrase in source language.
F2.6.6	**Use button**	Inserts phrase into composer. Inserts source text if card.sourceLang matches session.myLang; otherwise inserts target text. Closes overlay. If not in chat screen, navigates to chat.

6. Voice Subsystem

6.1 Text-to-Speech (TTS)

TTS uses Web Speech API SpeechSynthesisUtterance. All language codes are mapped to full BCP-47 locales before use.

en	th	ja	ko	zh	fr	de
en-US	th-TH	ja-JP	ko-KR	zh-CN	fr-FR	de-DE
es	**it**	**pt**	**ru**	**ar**	**vi**	**id**
es-ES	it-IT	pt-BR	ru-RU	ar-SA	vi-VN	id-ID

ms	fil	nl	sv	pl	tr
ms-MY	fil-PH	nl-NL	sv-SE	pl-PL	tr-TR

speak(text, lang, btn) cancels any active utterance before starting a new one. Clicking the active TTS button cancels speech. The .playing CSS class is toggled on the button element during playback.

6.2 Speech-to-Text (STT)

STT uses Web Speech API SpeechRecognition (webkitSpeechRecognition on Safari/iOS). If the API is unavailable, the Voice button is hidden. Recognition language is set to ttsLocale(session.myLang). Interim results populate the textarea live; final result replaces them. The button turns red with .active class during recognition.

7. Settings Screen

ID	Element / Action	Behaviour & Rules
F7.1	**Back navigation**	Returns to the screen the user came from (chat screen if opened from ··· menu; home screen otherwise). Origin screen stored in state._settingsOrigin.
F7.2	**Display name**	Text input. Saves to prefs.userName on blur. Also updates the Home screen My Name field.
F7.3	**You speak selector**	Updates prefs.myLang. If a session is active, also updates session.myLang and shows toast "Languages updated for this session".
F7.4	**Partner speaks selector**	Updates prefs.partnerLang. If a session is active, also updates session.partnerLang.
F7.5	**Translation engine info**	Read-only card. Green: "Chrome Built-in AI — works offline." Grey: "MyMemory API — internet required."
F7.6	**Saved conversations list**	Lists all sessions with message count and language pair. Each row has ↗ (reopen) and × (delete) buttons.
F7.7	**Phrasebook stats**	Card count and storage usage in KB.
F7.8	**About section**	App name, version (v1.7), brief description.

8. Clarify Subsystem

The clarify system allows a conversation participant to ask for clarification of a message they received, and allows the sender to provide a contextual explanation. Clarify state is per-message and persisted to storage.

ID	Element / Action	Behaviour & Rules
F8.1	**Request (receiving side)**	User taps "? Clarify" on a partner message. OVL-03 opens showing the quoted original text and an optional note textarea. Tapping "Ask →" sends {type:"clarify", id, note} over P2P and updates msg.clarify to

		{status:"requested", text:note}. Button label changes to "? Clarify" with c-active styling.
F8.2	**Notification (sending side)**	When a clarify packet is received, updateBblClarify() fires on the referenced message. The footer clarify anchor element becomes a visible "! Respond" button (amber, c-active). The detail pane auto-expands to show the clarification request text.
F8.3	**Answer (sending side)**	User taps "! Respond". OVL-04 opens showing quoted text + answer textarea with Markdown hint. Tapping "Send →" sends {type:"clarify-ans", id, answer}. msg.clarify updated to {status:"answered", text:answer}.
F8.4	**Display (receiving side)**	On receiving clarify-ans, updateBblClarify() fires. Button label changes to "✓ Clarified". Detail pane auto-expands and the clarify block shows the answer text (Markdown-rendered).
F8.5	**Wire format**	{type:"clarify", id:msgId, note:string} and {type:"clarify-ans", id:msgId, answer:string}.
F8.6	**Persistence**	clarify state is saved to the message in storage. On session reload, updateBblClarify() restores the visual state for each message with status !== "none".

9. Design Decisions (D-series)

ID	Decision	Rationale
D-01	**PeerJS for P2P transport**	Provides simple WebRTC abstraction with free public signalling. No server cost. Acceptable for the use case; users understand it requires internet for initial handshake.
D-02	**Session ID derived from names**	Makes session IDs human-readable and memorable. Trade-off: two different people with the same name+partner combination share an ID. Acceptable given the informal use case.
D-03	**MyMemory as cloud translation fallback**	Free tier supports the expected casual use volume. No API key required. Response format validated; MYMEMORY error strings are caught and treated as failure.
D-04	**TTS via Web Speech API only**	No external TTS API. Quality varies by browser/OS voice selection. Acceptable trade-off for zero-cost, zero-configuration audio.

D-05	**Back-translation as quality signal**	Rather than a confidence score (unavailable from free APIs), back-translation gives users a human-readable check. Verdict stored locally and displayed as a pill.
D-06	**Footer bar: Clarify + Save + ▾ only**	Verify ↺ moved inside the detail pane to reduce visual noise during live conversation. The 3-button bar is the minimal persistent surface needed for core actions.
D-07	**No greyed Clarify on own messages**	An inert placeholder button adds visual clutter with zero function. The "! Respond" button appears only when there is actually a clarification request to respond to.
D-08	**Phrase picker language-pair filter on by default**	When in an active session, showing all languages by default creates noise. Users can toggle "All langs" if needed.
D-09	**insertPhrase inserts source when sourceLang === myLang**	The user wants to send the phrase in the language they speak. If they stored "Hello → สวัสดี" and they speak English, inserting "Hello" is correct.
D-10	**endSessionAndSave uses callback pattern**	endSession() is passed as the onComplete callback to openBulkSave(). This ensures state is only cleared after the user has confirmed saving (or immediately if there is nothing to save).
D-11	**Settings back navigation uses origin screen**	goToSettings() stores state.screen in state._settingsOrigin. backFromSettings() restores it. A user who opens Settings from within a live chat returns to that chat.
D-12	**Join screen has I Speak language selector**	Language is set at join time, not inferred or forced by the host. The joining user's myLang is transmitted to the host via the hello handshake, updating the host's view of partnerLang.
D-13	**No automatic share on session start**	Auto-copying the clipboard on quickStart() caused confusion and failed silently on some mobile browsers. Share is explicit: tap the share icon in the session ID row.
D-14	**Card catalog layer uses immediate-save select**	A Save button for a single-field select adds unnecessary steps. onchange saves immediately and shows a toast. This matches the behaviour of all other layer selects in the app.

D-15	**Catalog layer shows only single catalog selection**	Cards support catalogIds as an array, but the UI presents a single-select. This keeps the interaction simple. Assigning to no catalog (empty option) puts the card in "All Phrases" only.

10. Visual Design Language

10.1 Colour Palette

Token	Hex	Usage
`--teal`	`#2E8B8B`	Primary action, active states, headers
`--teal-mid`	`#3A9E9E`	Hover border for cards
`--teal-light`	`#E6F4F4`	Teal background tint, chip backgrounds
`--ink`	`#1A1714`	Primary text
`--ink-mid`	`#5A5552`	Secondary text, icons
`--ink-dim`	`#9A9592`	Labels, placeholders, timestamps
`--border`	`#E2DDD8`	Card and input borders
`--paper`	`#F5F1EC`	Input backgrounds, card hover bg
`--cream`	`#FDFAF7`	App background
`--green`	`#2E7D4F`	Verified / correct state
`--amber`	`#C07830`	Warning, queued, clarify-request state
`--red`	`#C0392B`	Error, delete, flagged state
`--tag-bg`	`#EEE8FF`	Tag pill background
`--tag-text`	`#6B4FBB`	Tag pill text

10.2 Typography

Body: DM Sans (Google Fonts). Headings / logo / card titles: Lora (Google Fonts). Monospace: system font (session IDs, tag slugs). Base size 15px. All weights: 400 (regular), 500 (medium), 600 (semibold), 700 (bold).

10.3 Animation

Bubbles animate in with fadeUp (translateY 8px → 0, opacity 0 → 1, 170ms). Overlays slide up with slideUp (translateY 36px → 0, opacity 0.7 → 1, 220ms). Screens fade with opacity transition (180ms). ▾ toggle rotates 180° (200ms). Shimmer loading animation for pending translations.

11. Acceptance Criteria

11.1 Conversation Flow

The following must all pass for the Talk flow to be considered complete:

- Two users on separate browser instances can connect using a shared URL and exchange messages that appear in both languages.
- Messages sent while disconnected are queued and delivered when reconnection occurs.
- The session list on Home screen renders without error for both empty and populated states.
- TTS buttons play audio for both source and target text in each bubble using the correct language locale.
- The Voice (STT) button populates the composer textarea with recognised speech.
- Clarify request and response cycle completes: "? Clarify" → OVL-03 → "! Respond" → OVL-04 → "✓ Clarified" state visible on both sides.
- Back-translation verify cycle completes: Verify ↺ in detail pane → back text appears → ✓/✗ verdict → pill in footer bar.
- Bulk save at session end correctly identifies only new phrases (not already in library) and saves them.
- Settings opened from within chat returns to chat on back press.
- "Save phrases & end" completes saving before navigating away from chat.

11.2 Phrasebook Flow

- New catalog overlay appears and creates a catalog correctly.
- New phrase can be created with auto-translation, tags, and catalog assignment.
- Phrase card detail layers all function: inline editing for source, target, notes; verify cycle for back-translation; tag add/remove; catalog reassignment.
- Phrase picker in chat filters to session language pair by default, with "All langs" toggle.
- Use button inserts source text when card.sourceLang matches session.myLang; otherwise inserts target text.
- Use button on a card with no active session shows toast and does not crash.
- Deleting a catalog moves its phrases to "All Phrases" (not deleting them).

12. Version History

Version	Date	Changes
v1.0	Feb 2026	Initial Talk + Say implementation from alpha source files.
v1.5	Feb 2026	Full rewrite with P2P, card detail layers, clarify subsystem, bulk save, v1.5 bubble redesign.
v1.6	Mar 2026	27 bug fixes: nested template literal crash, getSides language logic, hello handshake, session pruning, verdict pills, 7 additional TTS locales, home card language selectors, sender name span.
v1.7	Mar 5 2026	15 fixes from full validation audit: ICON_TTS/ICON_SHARE undefined, renderSessionList crash (validSs), new-cat-overlay missing HTML, .bbl-sender CSS missing, endSessionAndSave state race, settings back navigation, join screen language selector, footer bar declutter (Verify moved to detail pane, inert Clarify span removed), STT button labelled, composer placeholder updated, phrase picker language filter + correct side insertion, auto-share removed from quickStart, Use button toast for no-session state, catalog reassignment layer in card detail, Markdown hint on clarify answer, version updated to v1.7.